

NLP-I Task: ICD-10 Codification

Deep Learning Baseline

Ernest Valveny (Ernest.Valveny@uab.cat)

Lei Kang (lkang@cvc.uab.cat)

Q2/1009

11/05/2026

Task Formulation

- Task type: **single-label multiclass classification**.
- Input: one clinical literal from `leaderboard_data.csv`.
- Output: one ICD category label, stored as `y_category`.
- Target definition:
 - `y_category = Code.astype(str).str[0]`
- Example:
 - `Code = "I420" -> y_category = "I"`
 - `Code = "3E03329" -> y_category = "3"`
- All categories:
 - `[0, 1, 2, ..., 9, a, b, ..., z]`

Data Pre-processing

Training file: codification_data.csv.

Test file: leaderboard_data.csv.

Required training columns: Code, Literal

Required test columns: id, Literal, y_category

Text preprocessing is intentionally light:

```
text = str(text)
```

```
text = text.strip()
```

```
text = re.sub(r"\s+", " ", text)
```

I do not lowercase text, not remove accents, not remove punctuation.

Because: I will use a pretrained tokenizer that can handle Spanish biomedical text directly.

Data Preparation

Build label classes from the training set: (number of classes: 36, [0, 1, 2, ..., 9, a, b, ..., z])

```
labels = sorted(train_df["y_category"].unique())  
label2id = {label: idx for idx, label in enumerate(labels)}  
id2label = {idx: label for label, idx in label2id.items()}
```

Train/validation split: train: validation == 80: 20, and split type: stratified, so class proportions are approximately preserved.

```
from sklearn.model_selection import train_test_split  
train_split, val_split = train_test_split(  
    train_df,  
    test_size=0.2,  
    random_state=args.seed,  
    stratify=train_df["label_id"],  
)
```



Backbone Model

[PlanTL-GOB-ES/roberta-base-biomedical-clinical-es](#)

Architecture: RoBERTa-base style encoder

pretrained with masked language modeling on Spanish biomedical and clinical corpora (more than 1B tokens, including biomedical crawled text, clinical cases, Spanish clinical notes/documents, SciELO, BARR2, Wikipedia life sciences, patents, EMEA, Medline, and PubMed.)

- Backbone type: roberta
- Transformer layers: 12
- Hidden feature size: 768
- Attention heads: 12
- Feed-forward intermediate size: 3072
- Number of parameters: 126M

PlanTL-GOB-ES/**roberta-base-biomedical-clinical-es**

Fill-Mask Transformers PyTorch Spanish roberta biomedical clini

Fine-tuning

Aggregation: <CLS> or Mean

- CLS aggregation: *features = hidden_states[:, 0, :]*
For RoBERTa, the first token is <s>, which plays the same role as BERT's [CLS].
- Mean pooling:
mask = attention_mask.unsqueeze(-1) # ignores padding tokens
*features = (hidden_states * mask).sum(dim=1) / mask.sum(dim=1)*

Classification Head

- A dropout layer followed by a linear layer:
Dropout(0.1)
Linear(768, 36)

Output shape:

- batch_size x num_classes
- Each row gets one predicted class.

Training Pipeline

1. Train on the training split.
2. Compute training loss (`nn.CrossEntropyLoss()`, `torch.optim.AdamW`).
3. Evaluate on the validation split.
4. Compute validation accuracy.
5. Save the best model based on validation accuracy.
6. Early stopping if validation accuracy does not improve for 10 epochs.

Hyperparameters	Value
Max epochs	50
Patience	10
Batch size	128
Learning rate	2e-5
Weight decay	0.01
Dropout	0.1

Results

leaderboard_data.csv

```
1 id,Literal,y_category
2 1,AMNIODRENAJE,0
3 2,Hiperparatiroidismo primario,E
4 3,MIGRANYA parto,0
5 4,VHC,B
6 5,Absceso mama izq,N
7 6,miomas parto,D
8 7,cos estrany,3
9 8,TCE,S
10 9,osteogénesis imperfecta,Q
11 10,Soplo sistólico,R
12 11,Amniorrexis,0
13 12,Tía materna + Ca mama,Z
14 13,laceraciones paraclitoriana,0
15 14,anti-D,3
16 15,J45.30,J
17 16,epoc reagudizado,J
18 17,Exprematura,V
19 18,asma parto,0
20 19,Alergia a Acaros,Z
21 20,alergia marisco,Z
22 21,Atrofia corticosubcortical,G
23 22,Trombopenia,D
24 23,HPB,6
25 24,Historia de RT,Z
26 25,Incontinencia urinaria de esfuerzo,R
27 26,ia,I
28 27,Hiperprolactinoma,2
29 28,COLESTASIS part,0
30 29,BSGC pélvica,0
```

10/05/2026

Method	Acc
<CLS>	0.565
Mean	0.570

#	Team	Members	Score	Entries	Last	Join
1	Group 0		0.570	2	1h	
 Your Best Entry! Your most recent submission scored 0.570, which is an improvement over your previous score of 0.565. Great job! Tweet this						
2	Eloi Mercader		0.569	3	9h	
3	Group 14	  	0.562	2	3h	
4	Group 10	  	0.548	1	4d	
5	GROUP 1		0.521	2	11m	
6	BAND	   	0.000	2	3d	
7	Eduard Pertinez		0.000	1	6h	

<https://www.kaggle.com/competitions/uab-asho-ai-codification>